Supporting Information

Development and Validation of a ReaxFF Reactive Force Field for Modeling Silicon–Carbon Composite Anode Materials in Lithium-Ion Batteries

Stéphane B. Olou'ou Guifo[a,d], Jonathan E. Mueller[a] *, Diana van Duin[b], Mahdi K. Talkhoncheh[b], Adri C. T. van Duin[b,c], David Henriques[d], Torsten Markus[d]

[a] Volkswagen Group, Berliner Ring 2, D-38436, Wolfsburg, Germany.

[b] RxFF Consulting LCC, 2398 Charleston Dr., State College, PA 16801, USA.

[c] Department of Mechanical Engineering, Pennsylvania State University, University Park, PA 16802, USA.

[d] Institute for Materials Science and Engineering, Mannheim University of Applied Sciences, Paul-Wittsack-Straße 10, D-68153, Mannheim, Germany.

* Corresponding author.
E-mail: jonathan.edward.mueller@volkswagen.de
Tel.: +49 5361 9 26875
Fax.: +49 5361 9 36627

A) ReaxFF force field parameters

Tab. S.1: General parameters.

Parameter	Unit	Value
$p_{boc,1}$	-	50.0000
$p_{boc,2}$	-	9.5469
$p_{coa,2}$	-	26.5405
$p_{trip,4}$	-	1.7224
$p_{trip,3}$	-	6.8702
kc_2	kcal mol^{-1}	60.4850
$p_{ovun,6}$	-	1.0588
$p_{trip,2}$	-	4.6000
$p_{ovun,7}$	-	12.1176
$p_{ovun,8}$	-	13.3056
$p_{trip,1}$	kcal mol^{-1}	–70.5044
$r_{Taper,lower}$	Å	0.0000
$r_{Taper,upper}$	Å	10.0000
$p_{val,7}$	-	33.8667
$p_{lp,1}$	-	6.0891
$p_{val,9}$	-	1.0563

Parameter	Unit	Value
$p_{val,10}$	-	2.0384
$p_{pen,2}$	-	6.9290
$p_{pen,3}$	-	0.3989
$p_{pen,4}$	-	3.9954
$p_{tor,2}$	-	5.7796
$p_{tor,3}$	-	10.0000
$p_{tor,4}$	-	1.9487
$p_{cot,2}$	-	2.1645
$p_{vdW,1}$	-	1.5591
B.O. cutoff	-	0.0010
$p_{coa,4}$	-	2.1365
$p_{ovun,4}$	-	0.6991
$p_{ovun,3}$	-	50.0000
$p_{val,8}$	-	1.8512
$p_{coa,3}$	-	2.6962

Tab. S.2: Atom parameters.

Parameter	Unit	Li	Si	C
r_o^{σ}	Å	2.1097	2.0057	1.4620
Val_i	-	1.0000	4.0000	4.0000
m	amu	6.9410	28.0600	12.0000
r_{vdW}	Å	2.1461	1.7098	2.0931
D_{ij}	kcal mol^{-1}	0.3726	0.2683	0.2002
γ	Å^{-1}	0.8651	0.3041	0.9751
r_o^{π}	Å	–0.1000	1.2635	1.2239
Val_i^{e}	-	1.0000	4.0000	4.0000
α	-	9.0000	12.5005	9.0000
γ_w	Å^{-1}	1.2063	5.2054	1.5000
Val_i^{angle}	-	1.0000	4.0000	4.0000
p_{ovun5}	kcal mol^{-1}	0.0000	21.7115	30.0000
χ	eV	–6.2351	3.8858	3.4759
η	eV	12.7757	5.3952	13.2596
$r_o^{\pi\pi}$	Å	–1.0000	–1.0000	1.1080
p_{lp2}	kcal mol^{-1}	0.0000	0.0000	0.0000
p_{boc4}	-	5.4409	11.7786	14.6589
p_{boc3}	-	6.9107	23.8188	24.4406
p_{boc5}	-	0.1973	0.8381	6.7313
p_{ovun2}	-	–17.1659	–2.5000	–8.2623
p_{val3}	-	2.2989	1.8326	5.0000
Val_i^{boc}	-	1.0000	4.0000	4.0000
p_{val5}	-	2.8103	2.5791	2.9663
$r_{core,2}$	Å	1.3000	1.4000	1.0000
$e_{core,2}$	-	0.2000	0.2000	0.1000
$a_{core,2}$	-	13.0000	13.0000	10.0000

Tab. S.3: Bond parameters.

Parameter	Unit	Li–Li	Si–Si	C–C	Si–Li	C–Li	Si–C
D_e^{σ}	kcal mol^{-1}	59.2876	71.2687	83.9626	26.4933	32.1300	109.7712
D_e^{π}	kcal mol^{-1}	0.0000	45.8533	134.4405	0.0000	–0.0200	0.0100
$D_e^{\pi\pi}$	kcal mol^{-1}	0.0000	17.1960	41.2045	0.0000	0.0000	0.0000
$p_{be,1}$	-	0.4363	–0.4953	0.7334	0.7248	0.9985	0.3939
$p_{bo,5}$	-	0.3000	–0.3182	–0.3932	0.3000	0.3000	–0.5558
13corr	-	0.0000	1.0000	1.0000	0.0000	0.0000	1.0000
$p_{bo,6}$	-	26.0000	15.9405	34.8447	26.0000	6.0000	17.2117
$p_{ovun,1}$	-	0.4590	0.0100	0.1823	0.0100	0.1936	0.1892
$p_{be,2}$	-	0.0485	0.0576	5.7939	0.7522	2.3857	0.4231
$p_{bo,3}$	-	0.0000	–0.4988	–0.2273	0.0000	–0.2500	–0.3104
$p_{bo,4}$	-	12.0000	7.2123	8.5764	12.0000	11.9965	7.2931
$p_{bo,1}$	-	–0.1990	–0.0846	–0.0898	–0.3406	–0.3191	–0.0677
$p_{bo,2}$	-	4.1914	5.7175	8.8434	6.4388	4.2036	6.3087
ovc	-	0.0000	0.0000	1.0000	0.0000	0.0000	1.0000

Tab. S.4: Off-diagonal terms.

Parameter	Unit	Si–Li	C–Li	Si–C
D_{ij}	kcal mol^{-1}	0.3183	0.3984	0.0707
R_{vdW}	Å	2.4157	2.1335	1.7011
α	-	9.0000	9.6455	12.5956
r_o^{σ}	Å	2.1443	1.5037	1.6610
r_o^{π}	Å	1.0000	1.0000	1.2804
$r_o^{\pi\pi}$	Å	1.0000	1.0000	–1.0000

Tab. S.5: Angle terms.

Parameter	Unit	Si–Si–Si	C–C–C	Si–C–C	C–Si–C	Si–C–Si	Si–Si–C
Θ_o	degree	74.0968	62.0574	2.5162	64.1836	90.0000	59.5681
$p_{val,1}$	kcal mol^{-1}	15.6349	16.1545	18.6626	39.9945	24.9953	14.6739
$p_{val,2}$	-	3.6545	3.0575	4.8139	1.3224	3.9887	8.0000
$p_{coa,1}$	kcal mol^{-1}	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
$p_{val,7}$	-	0.2525	0.0100	0.0775	0.0436	0.4129	2.9067
$p_{pen,1}$	-	0.0000	0.3556	0.0000	0.0000	0.0000	0.0000
$p_{val,4}$	-	1.2262	1.0000	1.0150	1.0920	1.2492	1.5730
Parameter	**Unit**	**Li–C–C**	**Si–C–Li**	**Li–Si–C**	**Si–Li–C**	**Si–Si–Li**	**-**
Θ_o	degree	82.8819	62.2920	79.9093	17.6669	76.2773	-
$p_{val,1}$	kcal mol^{-1}	4.6275	10.4389	13.5148	0.2335	16.7449	-
$p_{val,2}$	-	0.4373	1.5139	6.3243	0.8729	0.8811	-
$p_{coa,1}$	kcal mol^{-1}	0.0000	0.0000	0.0000	0.0000	0.0000	-
$p_{val,7}$	-	0.0100	2.5881	0.0100	0.0620	0.0161	-
$p_{pen,1}$	-	0.0000	0.0000	0.0000	0.0000	0.0000	-
$p_{val,4}$	-	2.0000	3.9806	1.5706	3.6834	1.0000	-

Tab. S.6: Torsion terms.

Parameter	Unit	C–C–C–C
V_1	kcal mol^{-1}	2.5000
V_2	kcal mol^{-1}	32.8238
V_3	kcal mol^{-1}	0.2056
$p_{tor,1}$	-	–9.0000
$p_{cot,1}$	kcal mol^{-1}	–2.2982

Force field version from January 12, 2022.

B) ReaxFF parameter training using *ab initio* data

Equilibrium states (ESs)

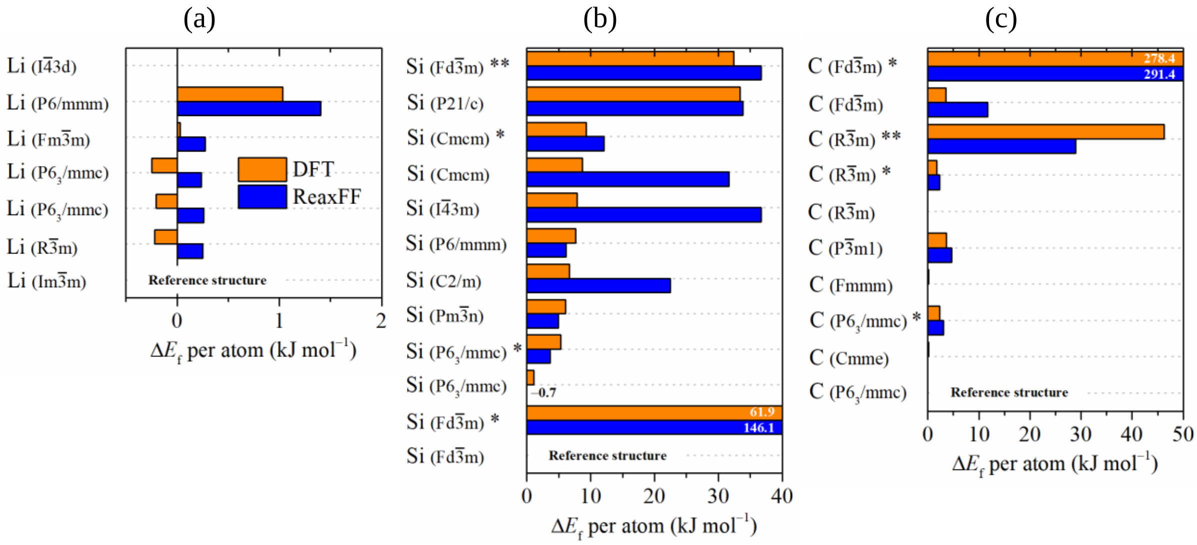

Fig. S.1: Formation energy of (a) Li, (b) Si and (c) C unary phases.

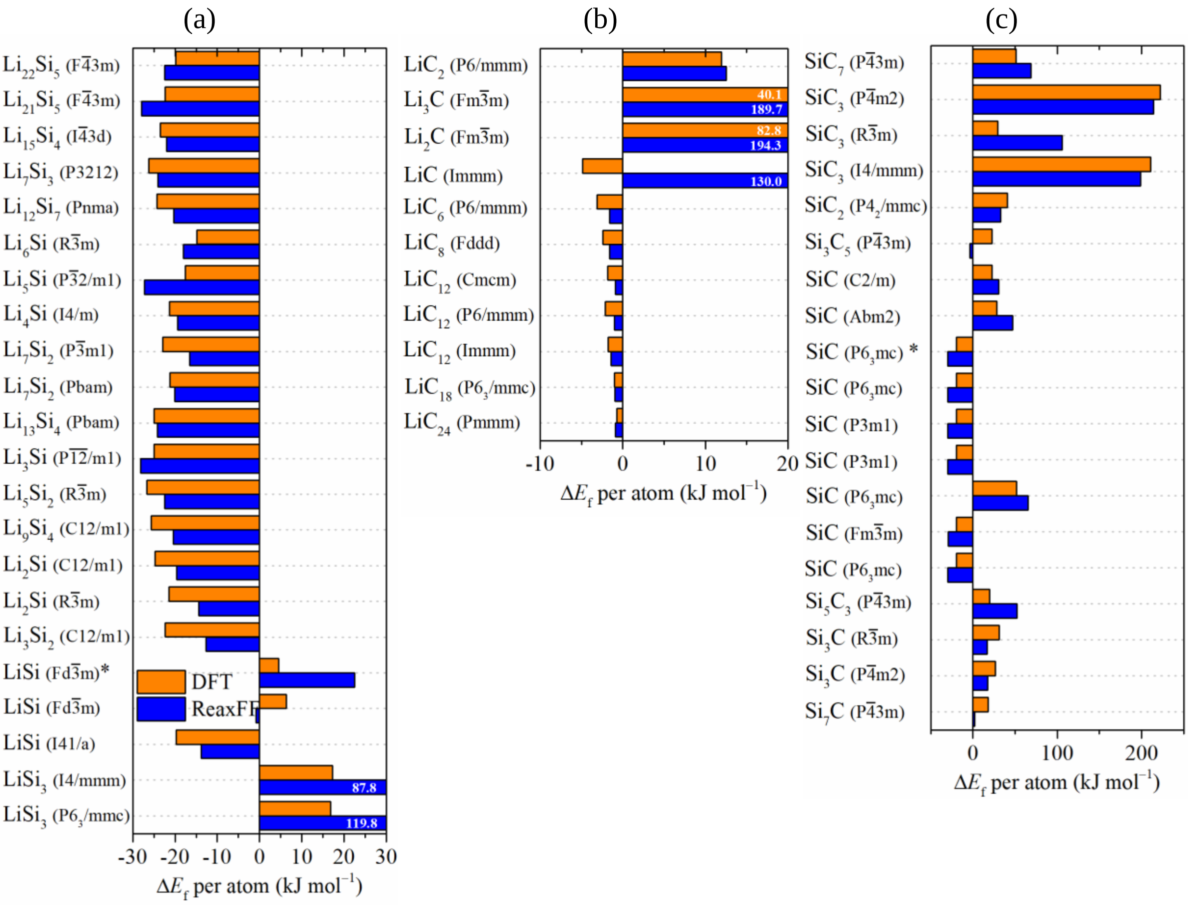

Fig. S.2: Formation energy of a) Li_xSi_y, (b) Li_xC_y and (c) Si_xC_y binary phases.

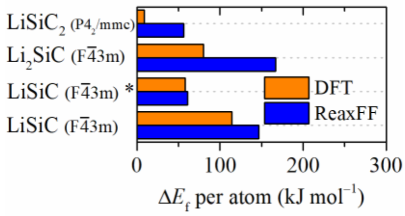

Fig. S.3: Formation energy of $Li_xSi_yC_z$ ternary phases.

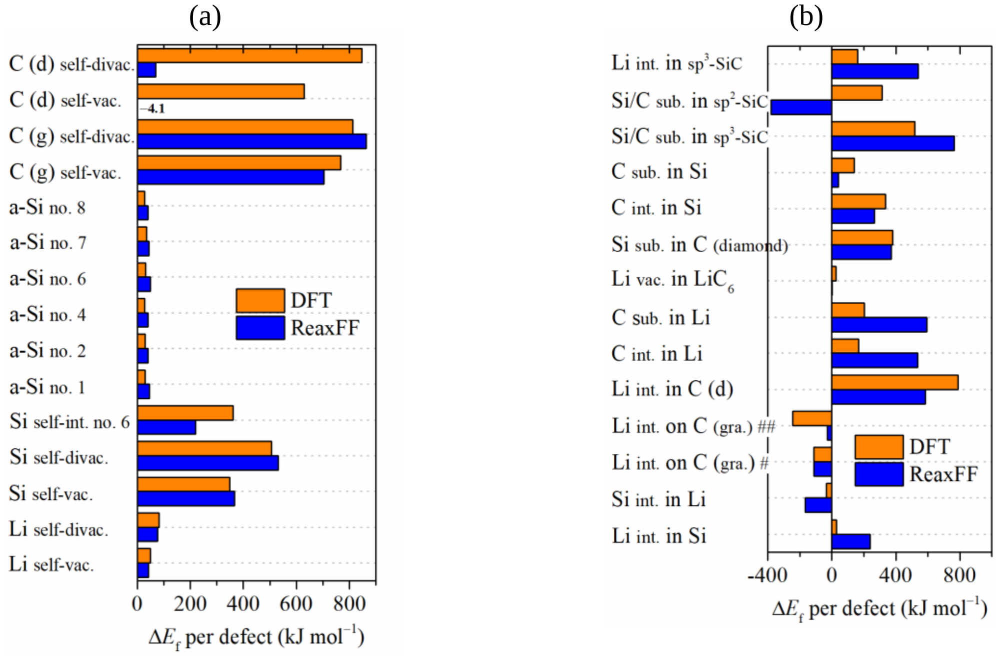

Fig. S.4: Formation energy of (a) self-defects and (b) foreign atoms.

vac., divac., int. and sub. stand for vacancy, divacancy, interstitial and substitutional defects, respectively. # and ## indicates a C vacancy and a C divacancy adjacent to the interstitial Li atom. sp^3-hybridized SiC is bulk silicon carbide (space group F$\bar{4}$3m) whereas sp^2-hybridized SiC forms a siligraphene sheet.

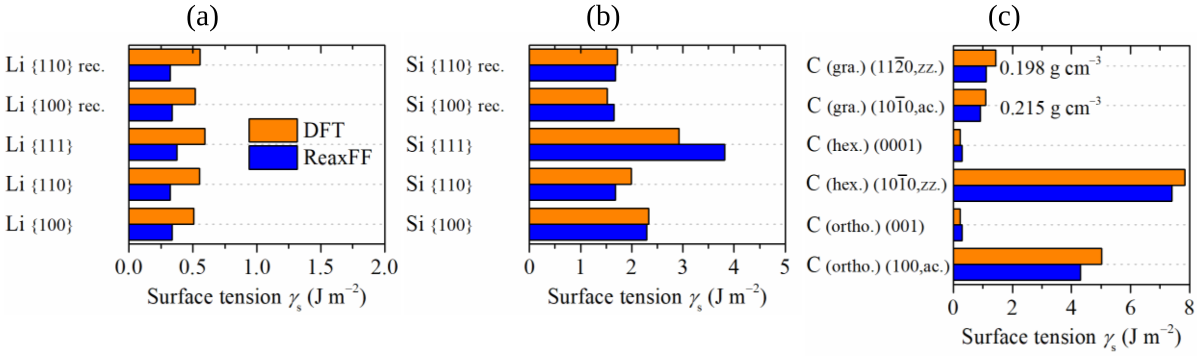

Fig. S.5: Surface tensions for unary phases built from (a) Li (Im$\bar{3}$m), (b) Si (Fd$\bar{3}$m) and (b) C (Cmme, P6$_3$/mmc and P6/mmm).

Reconstructed (rec.) surfaces in the training set were obtained from *ab initio* MD equilibration at 800 K for 200 fs with a time step of 2 fs. The surface tension of graphene layers with zigzag (zz.) and armchair (ac.) terminations is referenced to their mass density in the respective cell boxes.

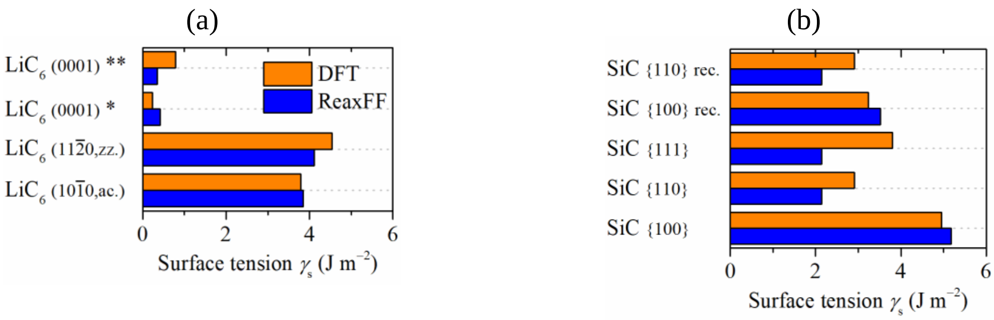

Fig. S.6: Surface tensions for binary phases from (a) LiC_6 (P6/mmm) and (b) SiC (F$\bar{4}$3m).

The suffixes * and ** indicate C and Li (adatoms) terminations on the LiC_6 slabs, respectively.

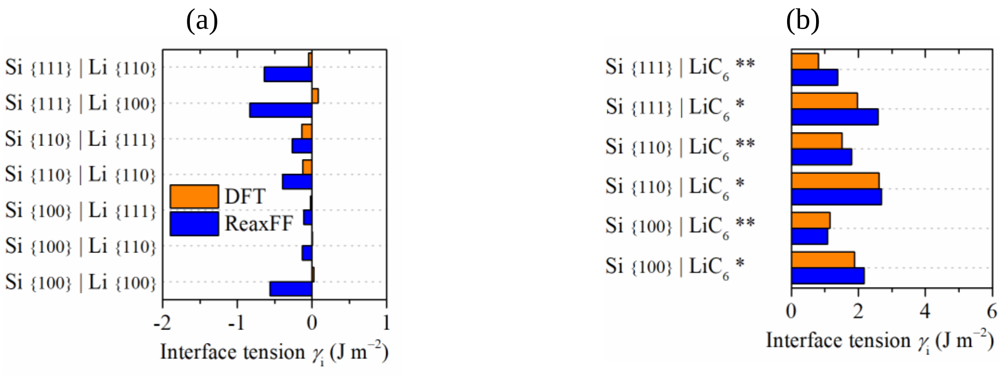

Fig. S.7: Interfacial tensions at (a) Si/Li and (b) Si/LiC_6 grain boundaries.

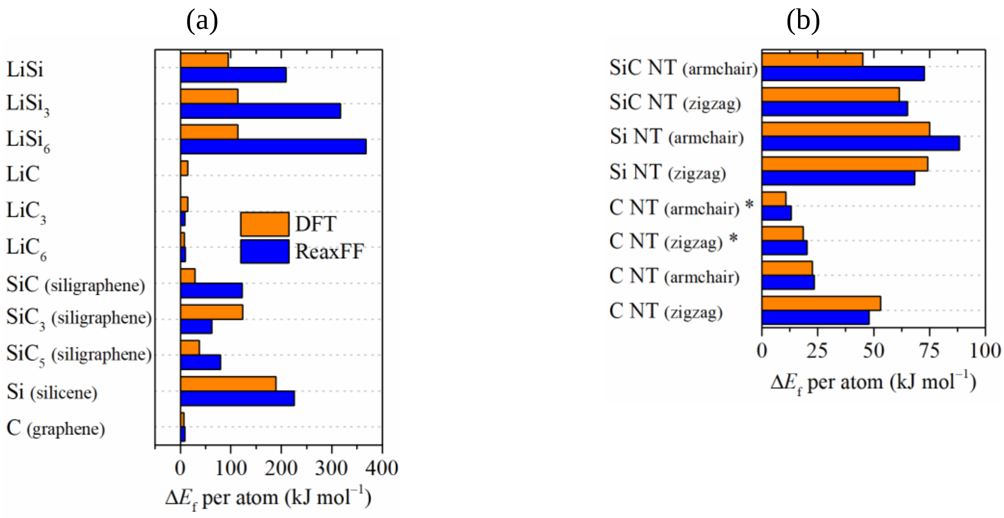

Fig. S.8: Formation energy in (a) nanosheets and (b) nanotubes (NTs) containing C, Si and/or Li.

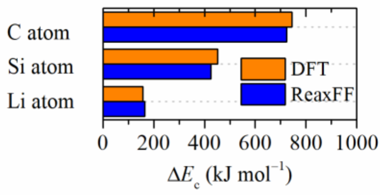

Fig. S.9: Cohesive energies of Li, Si and C with respect to their stable unary bulk crystals.

Equations of state (EoSs)

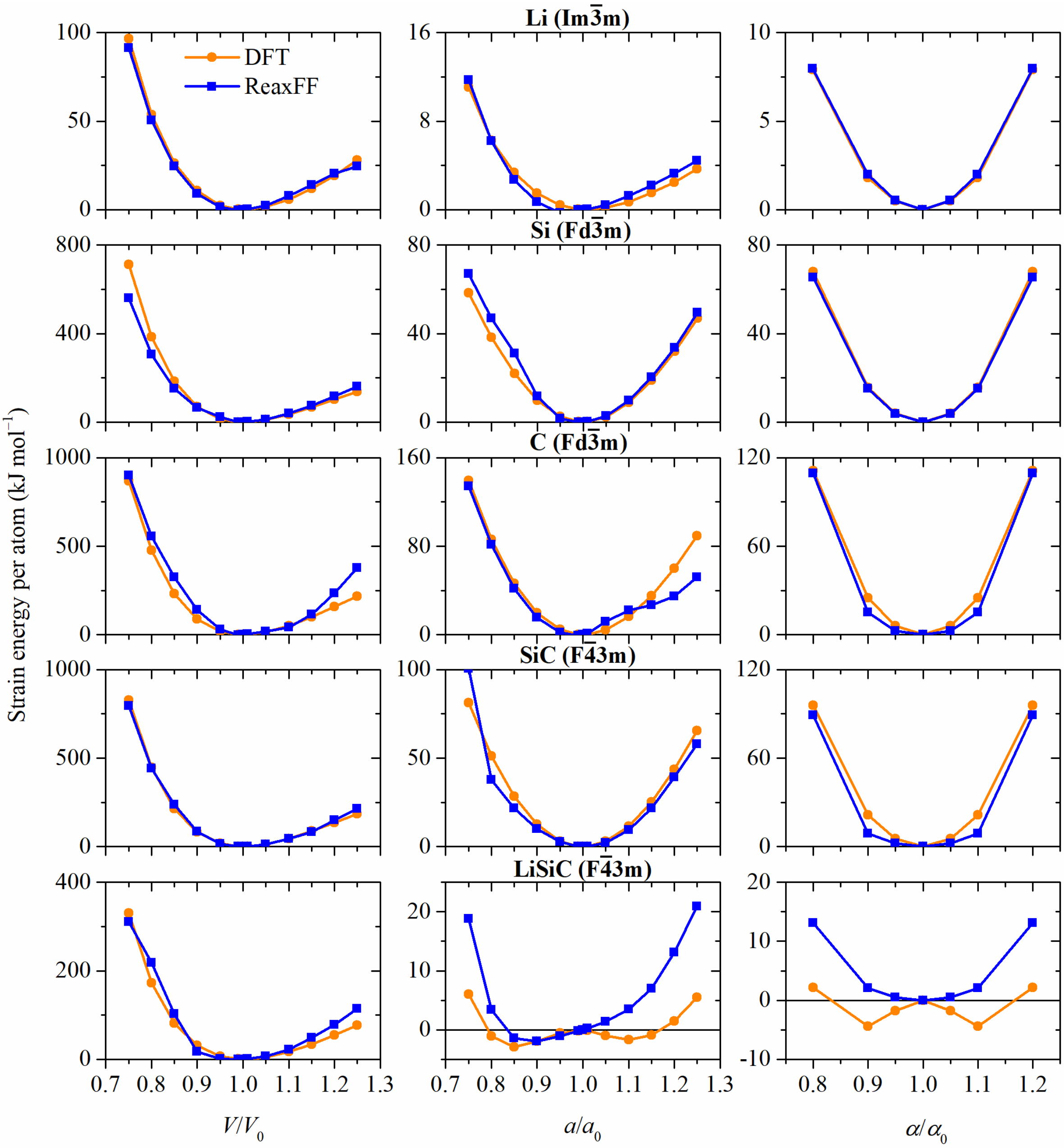

Fig. S.10: EoSs for cubic crystal structures.

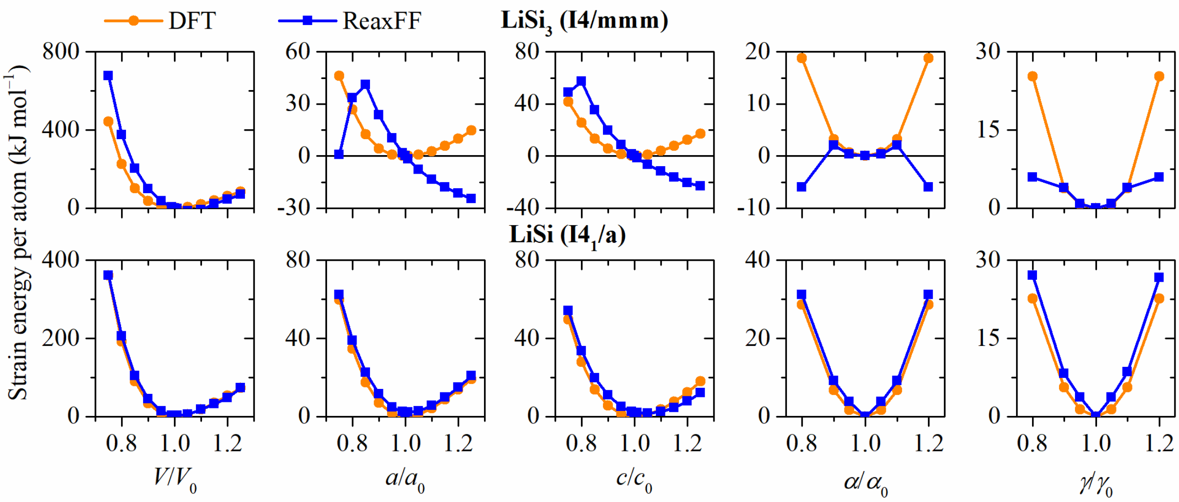

Fig. S.11: EoSs for tetragonal crystal structures.

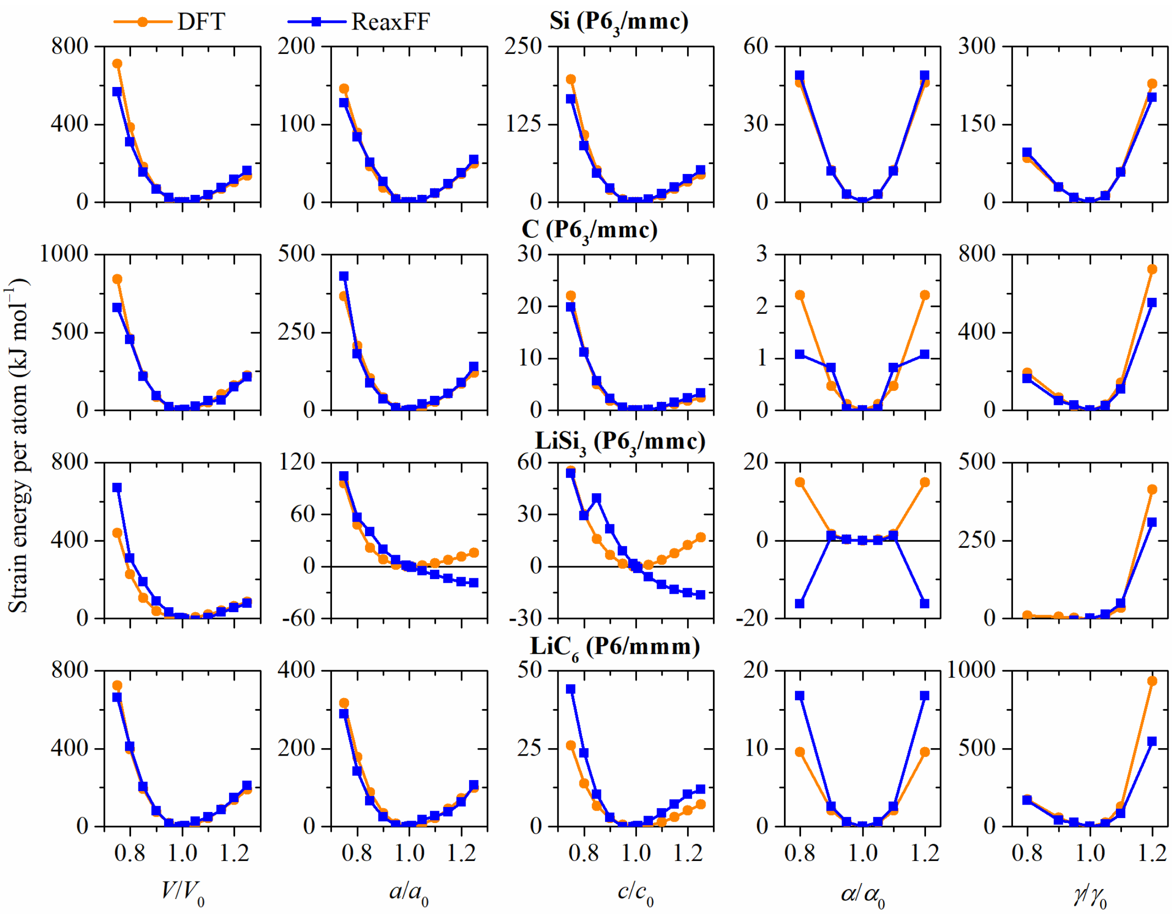

Fig. S.12: EoSs for hexagonal crystal structures.

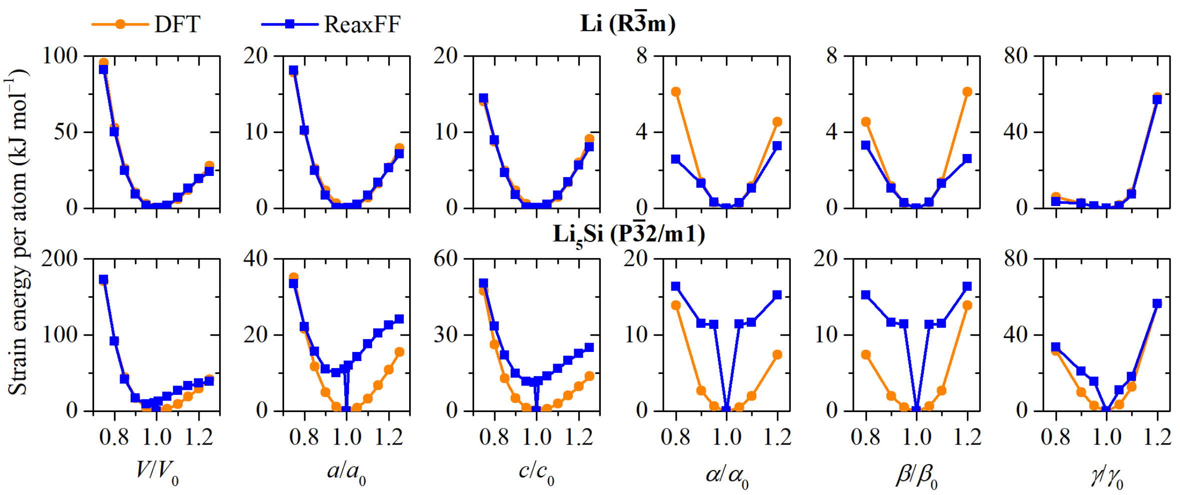

Fig. S.13: EoSs for rhombohedral (trigonal) crystal structures.

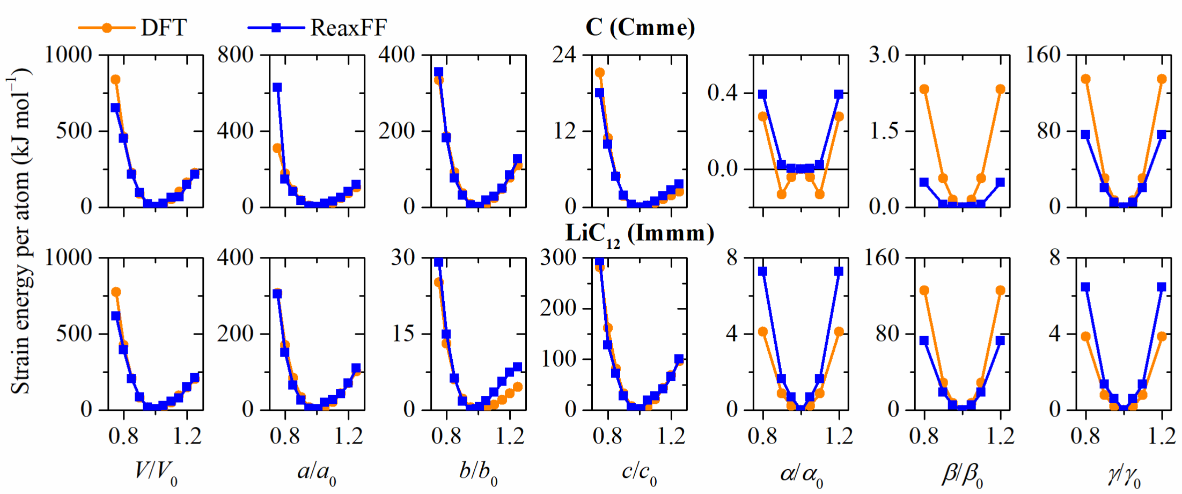

Fig. S.14: EoSs for orthorhombic crystal structures.

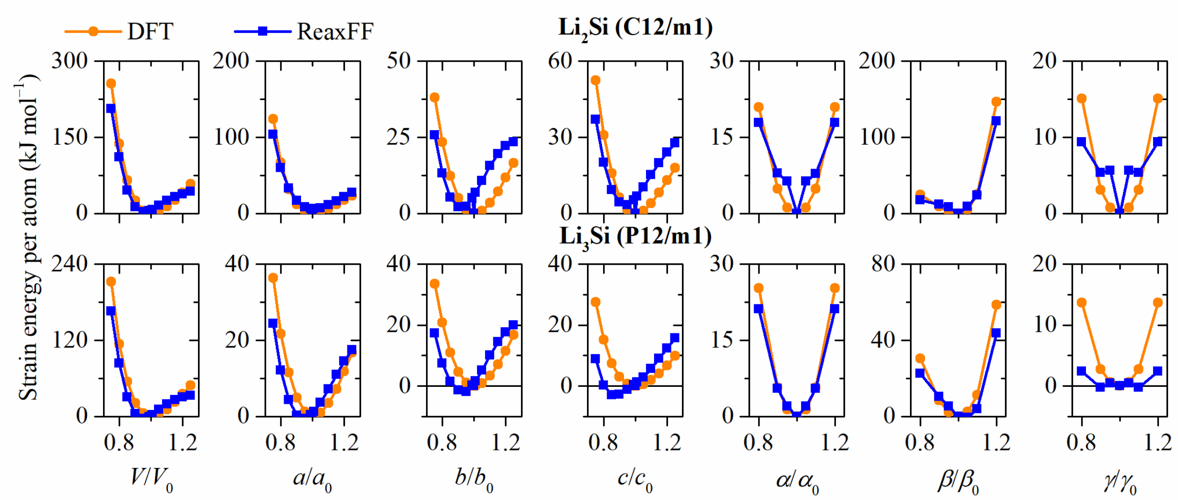

Fig. S.15: EoSs for monoclinic crystal structures.

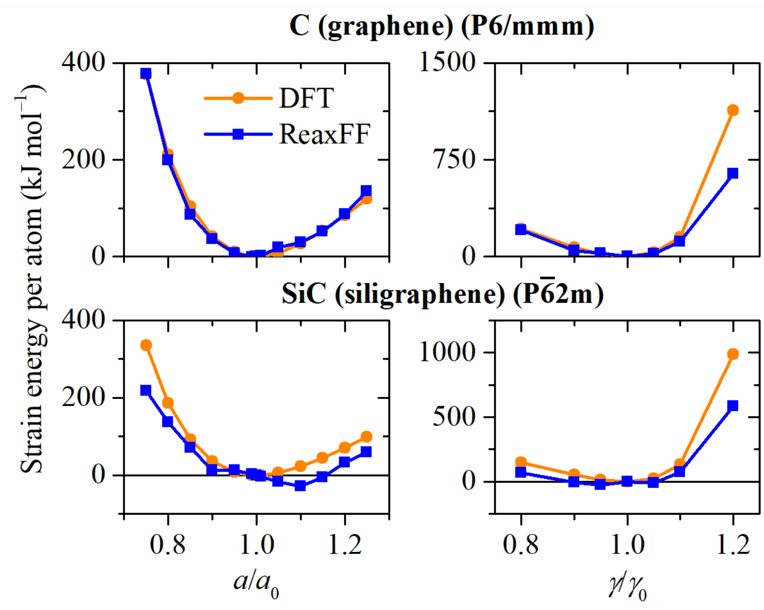

Fig. S.16: EoSs for single nanosheets against vacuum.

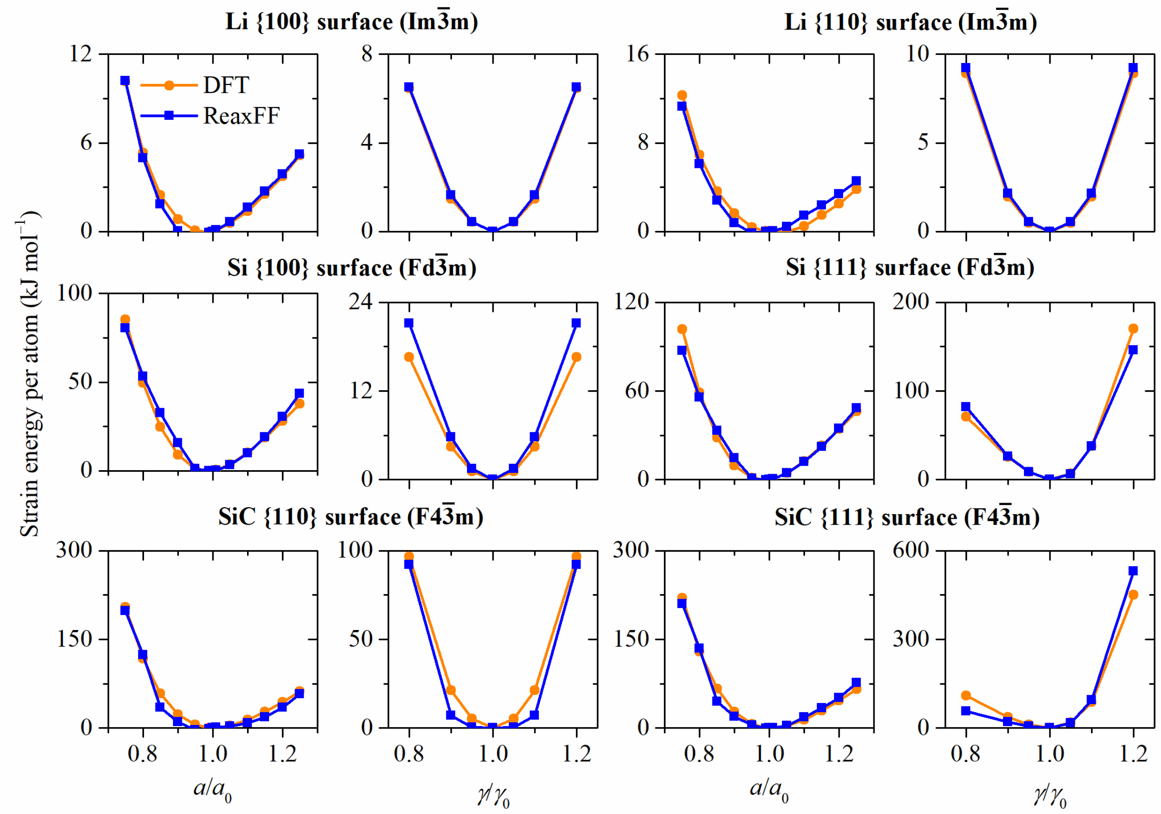

Fig. S.17: EoSs for surface slabs.

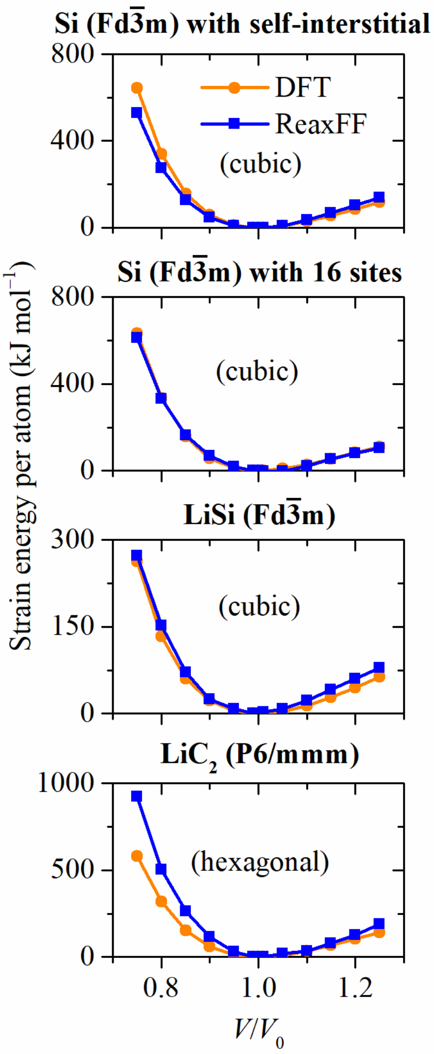

Fig. S.18: EoSs for additional crystal structures.

Minimum energy paths (MEPs)

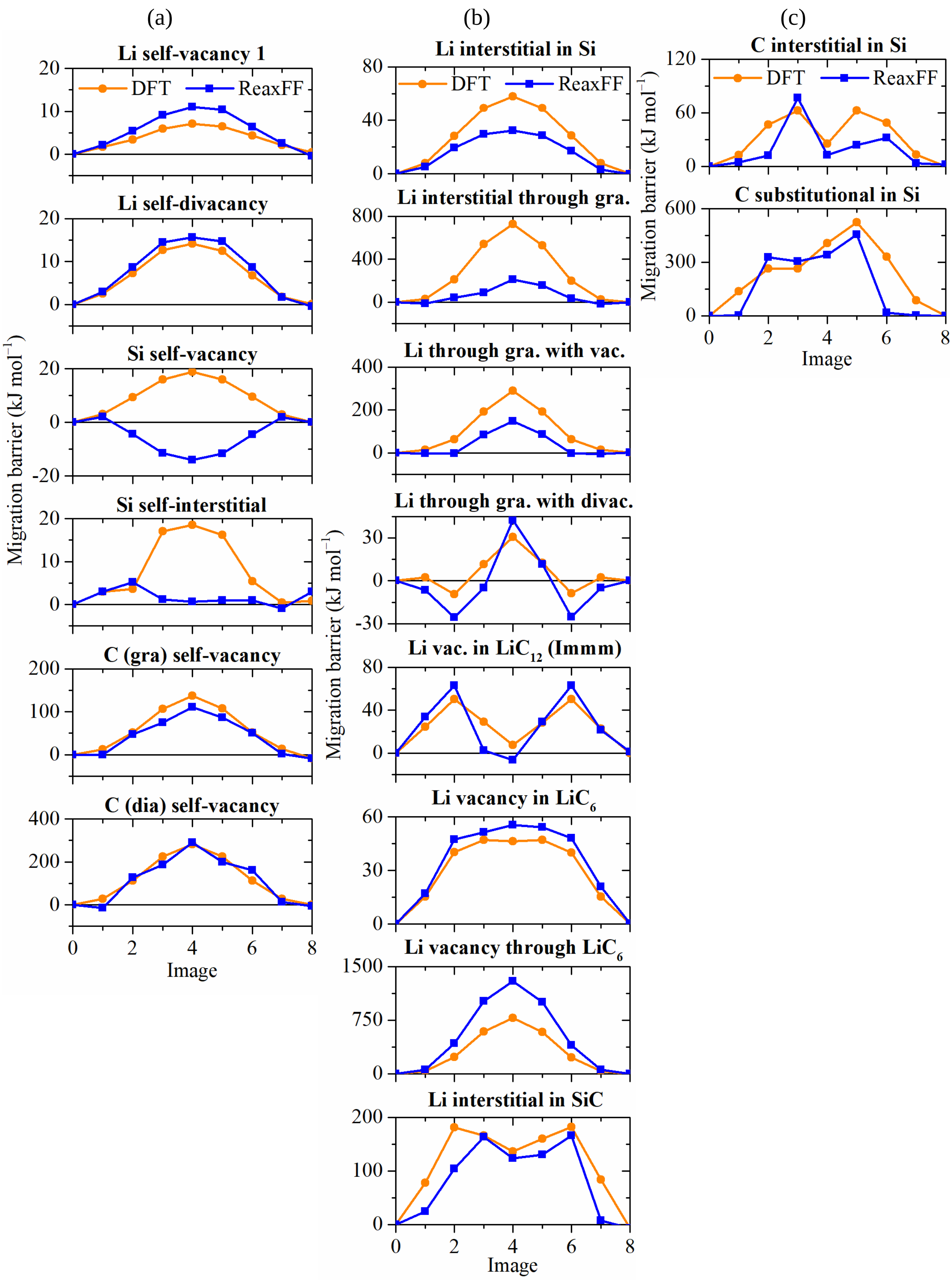

Fig. S.19: Minimum energy paths (MEPs) from ReaxFF and DFT training data for (a) self-diffusion, (b) Li diffusion in host structures and (c) C diffusion in Si.

C) ReaxFF parameter validation using *ab initio* data

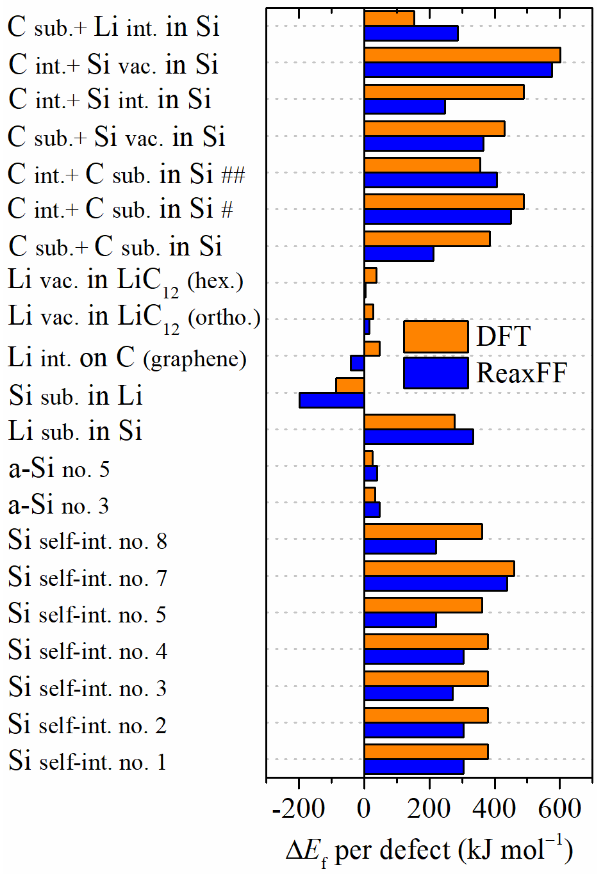

Fig. S.20: Formation energies of point defects (ReaxFF validation).

The [C int., C sub.] (int.: interstitial; sub.: substitutional) complexes in Si with suffixes # and ## are characterized by a single and a double bond between the C defects, respectively. hex. and ortho. stand for hexagonal and orthogonal crystal symmetries.

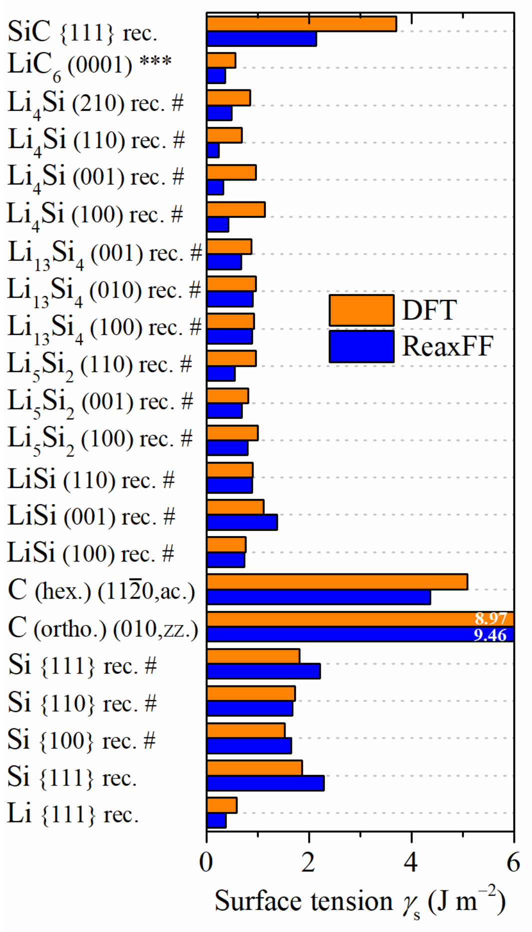

Fig. S.21: Surface tensions of reconstructed Li_xSi_y and graphitic slabs (ReaxFF validation).

The suffix * indicates the opposite C and Li (adatoms) terminations of the LiC_6 slab. # Additional Li_xSi_y surfaces were obtained via annealing at 500 K for 400 fs with a time step of 2 fs. The hexagonal C 11$\bar{2}$0 surface and orthorhombic C 010 surface have armchair (ac.) and zigzag (zz.) terminations, respectively.**

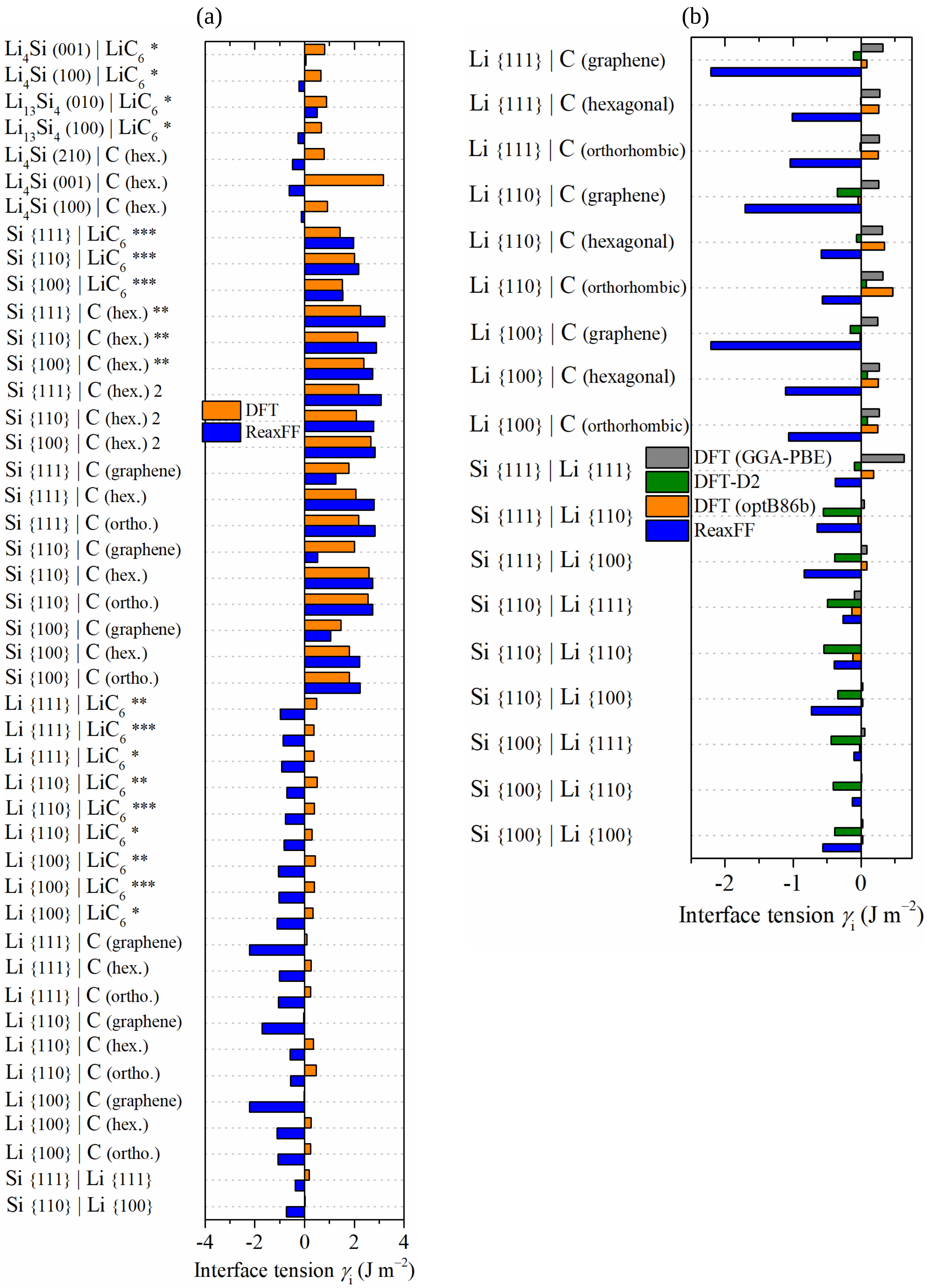

Fig. S.22: (a) Interfacial tensions (ReaxFF validation) and (b) comparison with other DFT models (DFT-D2 Grimme van der Waals force field correction and uncorrected DFT using GGA-PBE functionals).

The symbols *, ** and * stand for C, Li and combined C/Li terminations of the respective LiC_6 slabs. ReaxFF parameters tend to stabilize interfaces with different Li chemical potentials (concentrations) against the respective bulk structures due to the heterogeneous Li charge distribution. Regardless of the structure, ReaxFF assigns charges around +0.3 e to ionic Li compared DFT Bader charges lying around +0.8 e, closer to empirical Li ions (Li^+).**

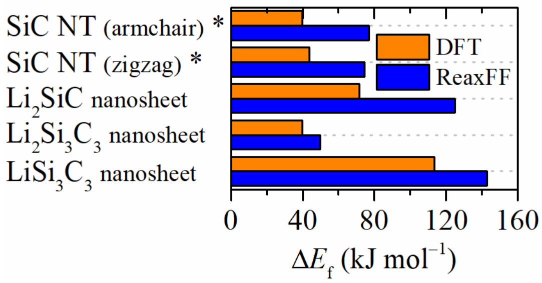

Fig. S.23: Formation energies of lithiated SiC nanosheets and delithiated SiC nanotubes (NTs) (ReaxFF validation).

Equations of state

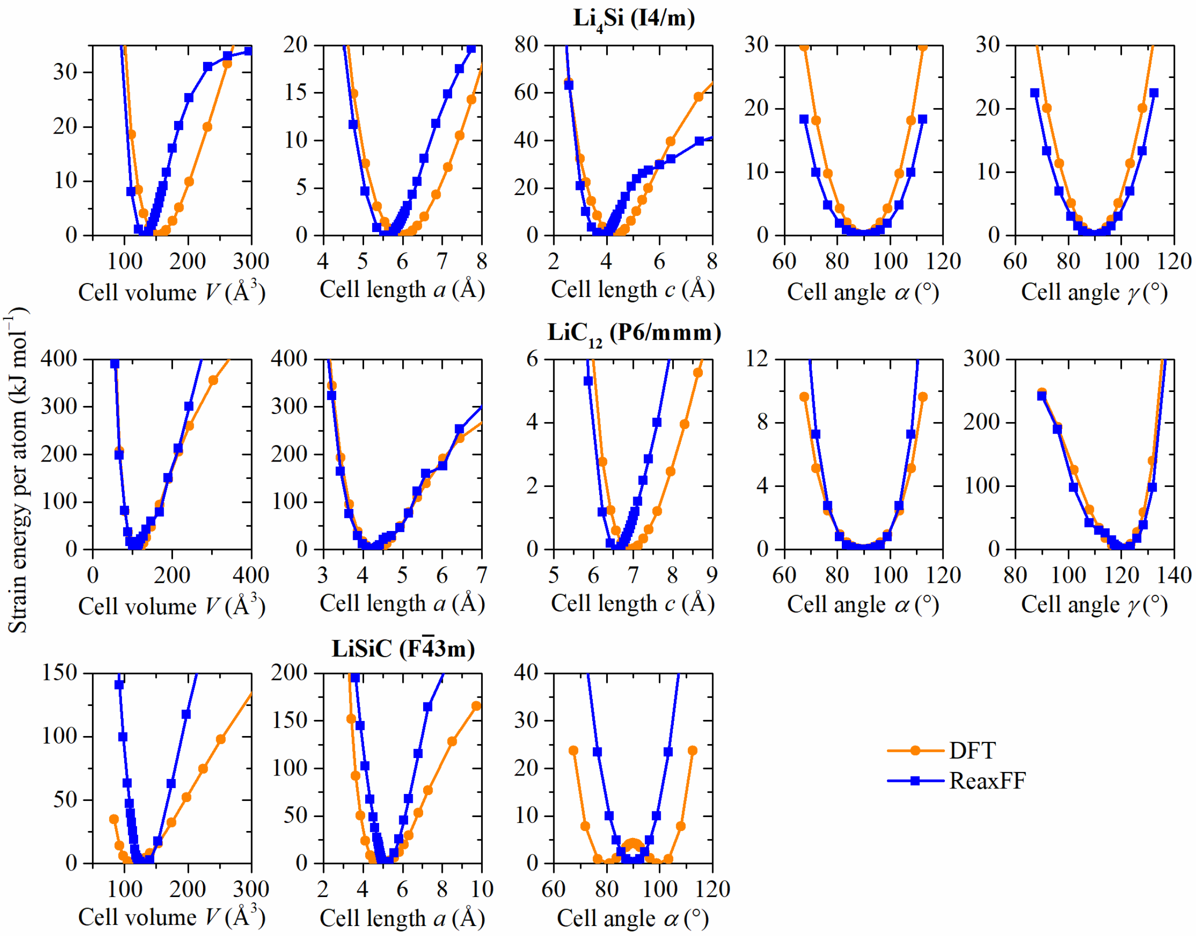

Fig. S.24: EoSs for bulk structures (ReaxFF validation).

Tetragonal Li_4Si, hexagonal LiC_{12} and cubic LiSiC.

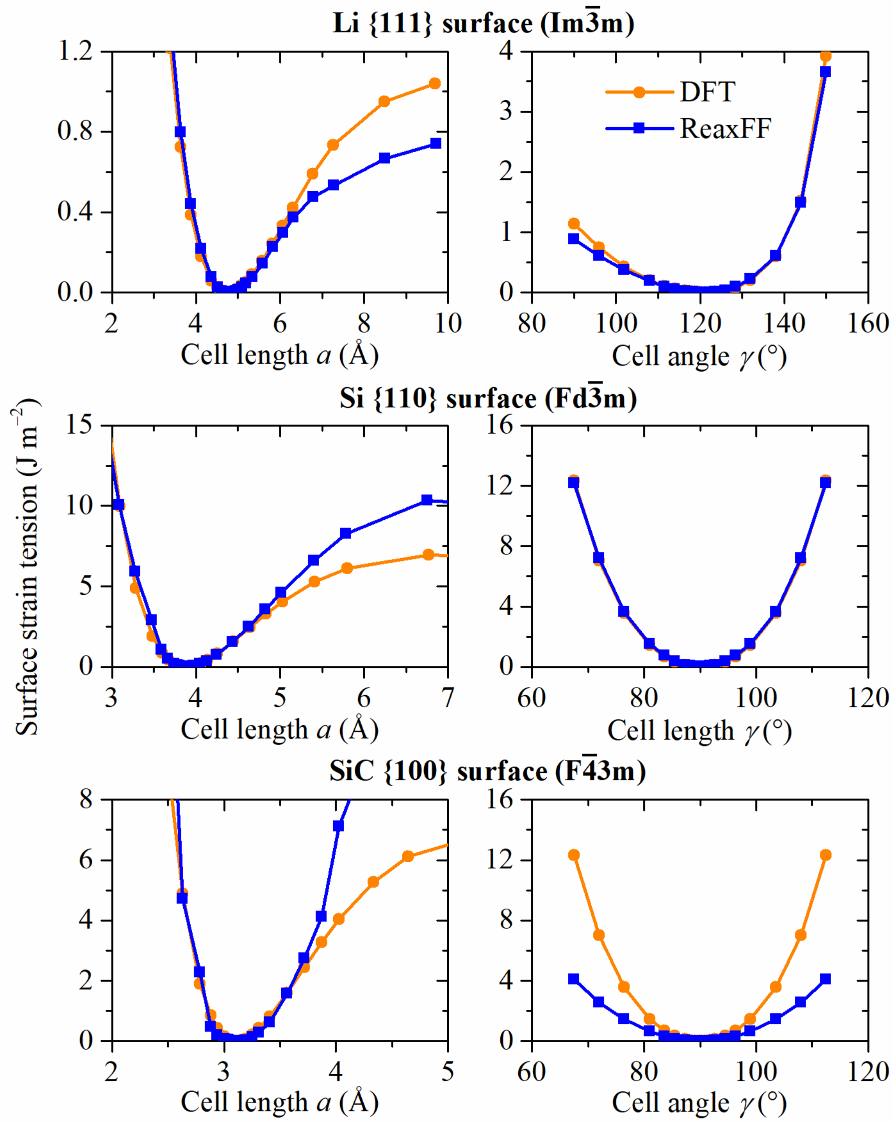

Fig. S.25: EoSs for surfaces (ReaxFF validation).

Li {1 1 1}, Si {1 1 0} and SiC {1 0 0} surfaces.

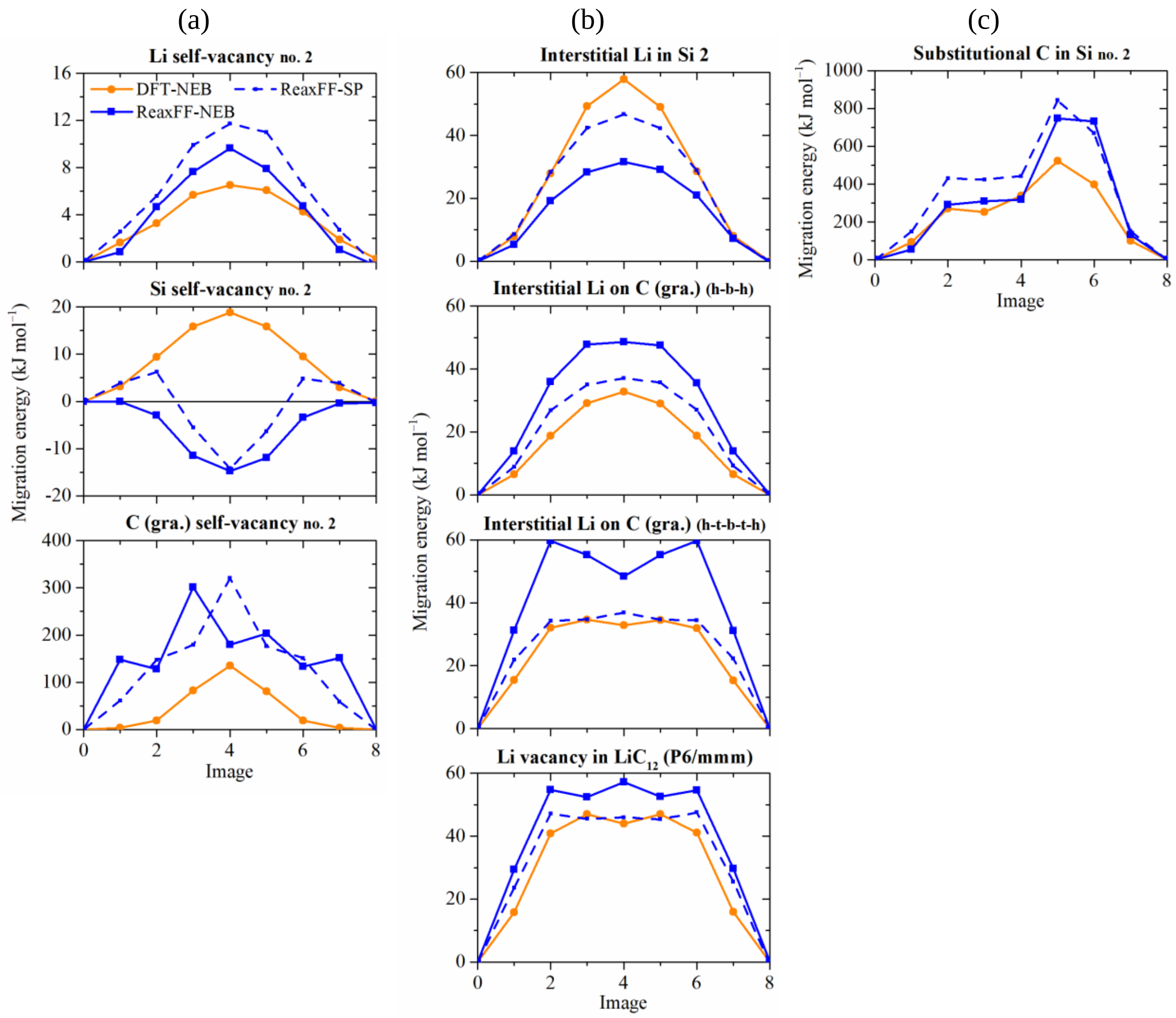

Fig. S.26: MEPs from ReaxFF and DFT validation data in conjunction with the NEB method for (a) self-diffusion, (b) Li diffusion in host structures and (c) C diffusion in Si (ReaxFF validation).

Transition states which were refined for each computed MEP are not explicitly shown because of the missing diffusion coordinates in the diagrams. ReaxFF single point (SP) results were computed directly from the *images* resulting from DFT-NEB optimization. h, t and b stand for hollow, top and bridge sites toward a single graphene (gra.) sheet.

D) Stability-density relationship of silicon carbide

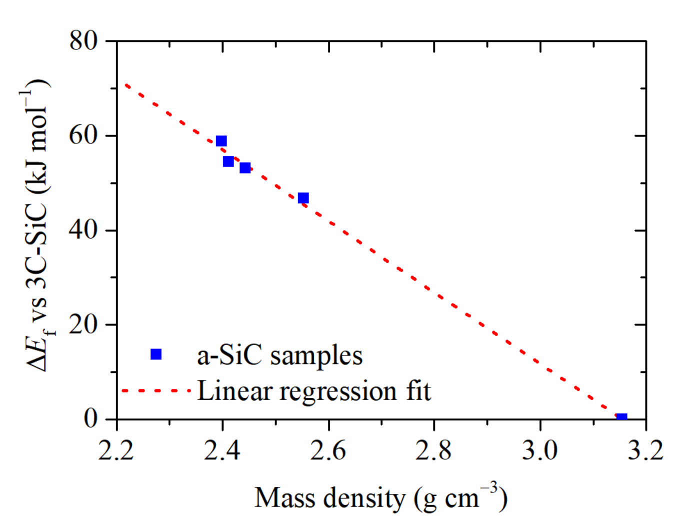

Fig. S.27: Stability of amorphous silicon carbide (a-SiC) compounds with respect to crystalline silicon carbides (c-SiC) as a function of mass density.

Depending on how it was synthesized c-SiC can crystallize into 3C-SiC (β-, $F\bar{4}3m$), 4H-SiC ($P6_3mmc$) or 6H-SiC (α-, $P6_3mmc$) with similar cohesive energies. Various methods estimate this cohesive energy to be: 738.1 kJ mol^{-1} (DFT), 600.7 kJ mol^{-1} (ReaxFF MD), and 611.7 kJ mol^{-1} (experiment[1]).

E) Lithiation at sp^2-C/a-Si grain boundaries

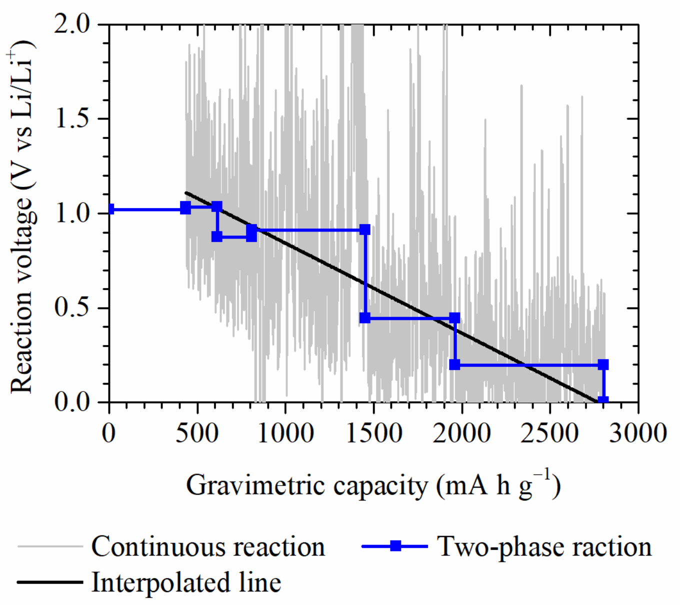

Fig. S.28: Lithiation voltage at Si/C grain boundaries.

The voltage is based on two-phase reactions involving consecutive lithiation steps obtained from GCMC simulations.

F) Computed physicochemical properties

Local chemical environment

The chemical environment of individual atoms or collections of atoms (*e.g.* all atoms of a given chemical species) can be analyzed using the radial distribution function (RDF), alternatively known as the pair correlation function:

$$g(r)=\frac{\rho(r)}{\rho_B}=\frac{dn(r)}{\rho_B 4\pi r^2 dr} \tag{S.1}$$

and the (radial) coordination number (CN)

$$CN(r_1)=4\pi\rho_B \int_0^{r_1} g(r) r^2 dr\,, \tag{S.2}$$

where ρ_B=N/V denotes the bulk density and N is the number of particles within the spherical volume $V=(4/3)\pi R^3$.[2] The first RDF peak corresponds to the coordination of the given atom(s) or specie(s).

Diffusion

Molecular diffusion can be treated in terms of mean-squared displacements (MSDs) based on atomic trajectories $\mathbf{r}_\alpha$ for respective atoms α:[3]

$$\langle R^2(t)\rangle=\frac{1}{N}\sum_\alpha |\mathbf{r}_\alpha(t)-\mathbf{r}_\alpha(t=0)|^2 \tag{S.3}$$

The tracer diffusion coefficient is then

$$D=\frac{1}{2d}\lim_{t\to\infty}\frac{d\langle R^2(t)\rangle}{dt}\,. \tag{S.4}$$

Diffusion coefficients were extracted from MD simulations performed within the NVE ensemble, on systems, which were first equilibrated in either the NPT ensemble (bulk, *i.e.* three-dimensional, systems) or the NVT ensemble (interfacial, *i.e.*, two-dimensional, systems). Indeed, as the volume and the total energy of a pre-equilibrated system are fixed it is possible to maintain the temperature (and pressure) from the previous NVT (or NPT) simulation fairly well, while evolving without artificial interference from a thermostat (and barostat) within the NVE ensemble. Subsequently, linear regression is applied within the interquartile range of the MSD curve, in order to determine the tracer diffusion coefficients at equilibrium. Since diffusion is a thermally activated mechanism it can be tracked more easily at high temperatures. Below their melting point, most materials exhibit a temperature dependence which is well approximated by the Arrhenius equation

$$D(T)=D_0 \exp\left(-\frac{Q}{RT}\right). \tag{S.5}$$

Hence, Li and host (inter-)diffusion coefficients at room temperature (RT), *i.e*. 25 °C, can be determined by fitting the activation energy Q and the pre-exponential frequency factor D_0 from high-temperature diffusivities by means of the least-square method.

Thermodynamic quantities

The Gibbs phase rule[4] explains the number of degrees of freedom of a system at isothermal-isobaric conditions:

$$f=C-P+N_{def}, \tag{S.6}$$

where C, P and N_{def} are the numbers of components, phases and independent defects such as surfaces and grain boundaries, respectively. The simulations considered in this work are targeted at standard temperature and pressure (STP), *i.e*., 25 °C and 1 atm. Hence for each atom, the mechanical energy contribution $P\Delta V$, at atmospheric pressure P is usually of the order of magnitude of 0.01 meV whereas the heat exchange $T\Delta S$ with the surroundings lies around 25 meV[5–9]. On the other hand, typical formation, diffusion or reaction energies ΔE *e.g*. involving Li and electrode materials such as Si or C lie in an order of magnitude of at least 100 meV.[10,11] Consequently, changes in both the Gibbs free energy, G, and Helmholtz free energy, A, characterizing the energetics of electrochemical reactions and thermodynamic equilibration in the NVT and NPT ensembles, respectively, are well approximated by differences in the enthalpy H or the internal energy E of the system, as given by the total energy computed with either the ReaxFF or DFT engine:

$$T\Delta S,P\Delta V\ll\Delta E \Rightarrow \Delta G\cong\Delta A\cong\Delta H\cong\Delta E. \tag{S.7}$$

The thermodynamic stability of an electrode as a function of the battery state of charge (SoC) can be investigated with respect to the delithiated host half-cell against body-centered cubic (bcc) Li. The goal here is to find out whether the host material would likely store Li ions by forming a Li_x[host] phase, *i.e*., with negative reaction (*i.e*. formation or mixing) energy. Regardless of the kinetic mechanism (*e.g*. intercalation, alloying or conversion), the reaction can be written as:

$$[\text{host}]+x\left(Li^{+}+e^{-}\right)\longrightarrow Li_x[\text{host}]. \tag{S.8}$$

The formation energy of the Li_x[host] phase is

$$\Delta E_f(x)=E_{Li_x[\text{host}]}-E_{[\text{host}]}-x\mu^0_{Li} < 0, \tag{S.9}$$

where μ^0_{Li} and $\mu^0_{[\text{host}]}$ are the chemical potential of pure, bcc Li and of the delithiated host structure, respectively. If the host material is composed of two or more components (we take Si and C as examples here), the thermodynamic stability with respect to single-components reference materials can be evaluated using:

$$\Delta E_f(x,y)=E_{Li_x[\text{host}]}-x\mu^0_{Li}-y\mu^0_{Si}-(1-y)\mu^0_{C}, \tag{S.10}$$

where the mole fractions for Si and C are y and 1–y, respectively.

Electrochemical properties

Electrochemical properties can be characterized thermodynamically by considering the following electrochemical reactions with a transfer of Δx moles of Li ions between reactant and product:[7,12]

$$\mathrm{Li_x[host]}+\Delta x\left(\mathrm{Li^+}+\mathrm{e^-}\right)\longrightarrow \mathrm{Li_{x+\Delta x}[host]}. \tag{S.11}$$

By formulating the chemical potential μ_{Li} of dilute Li-ions as

$$\mu_{\mathrm{Li}}(x)=\frac{\partial E_{\mathrm{Li_x[host]}}}{\partial x}, \tag{S.12}$$

the OCV is therefore determined from the Nernst equation as:[13]

$$V_{\mathrm{oc}}(x)=-\frac{1}{z_{\mathrm{Li}}F}\left(\mu_{\mathrm{Li}}(x)-\mu_0\right)=-\frac{E_{\mathrm{Li_{x+\Delta x}[host]}}-E_{\mathrm{Li_x[host]}}-\Delta x\mu_0}{z_{\mathrm{Li}}F\Delta x}. \tag{S.13}$$

A $\mathrm{Li_x[host]}$ phase with a positive voltage versus metallic Li has a Li storage capacity, which can be expressed gravimetrically as:

$$Q_{\mathrm{th,m}}(x)=\frac{z_{\mathrm{Li}}Fx}{M_{\mathrm{Li_x[host]}}}, \tag{S.14}$$

or volumetrically as:

$$Q_{\mathrm{th,V}}(x)=\frac{z_{\mathrm{Li}}Fx}{V_{\mathrm{Li_x[host]}}}=Q_{\mathrm{th,m}}\rho_{\mathrm{Li_x[host]}}. \tag{S.15}$$

In the case of a heterogeneous host structure composed of Si and C (by way of example), the total theoretical capacity of $\mathrm{Si_yC_{1-y}}$ is:

$$Q_{\mathrm{th}}=yQ_{\mathrm{th,Si}}+(1-y)Q_{\mathrm{th,C}}. \tag{S.16}$$

Surface and interfacial tensions

The tension of a homogeneous slab with a surface ω bounded by a vacuum region against its bulk phase ϕ is given by[14,15]

$$\gamma_{\omega}^{\phi}=\frac{1}{2A_{\omega}^{\phi}}\left(E_{\omega}^{\phi}-E^{\phi}\right), \tag{S.17}$$

where E_{ω}^{ϕ} and E^{ϕ} are the energies of the slab and of its bulk phase and A_{ω}^{ϕ} is the surface area of one side of the slab. The stability of solid/solid interfaces or grain boundaries between two phases ϕ_1 and ϕ_2 with the respective surface orientations ω_1 and ω_2 can be computed by calculating the interfacial tension as follows:[14,16]

$$\gamma_{\omega_1\omega_2}^{\phi_1/\phi_2}=\frac{1}{2A_{\omega_1\omega_2}^{\phi_1/\phi_2}}\left(E_{\omega_1\omega_2}^{\phi_1/\phi_2}-E^{\phi_1}-E^{\phi_2}\right), \tag{S.18}$$

where $E_{\omega_1\omega_2}^{\phi_1/\phi_2}$ is the energy of the two-phase system containing both interfaces and E^{ϕ_1} and E^{ϕ_2} are the energies of the respective bulk phases without defects.

Defect energies and concentrations

The formation free enthalpy of point defects such as vacancies and interstitially and substitutionally dissolved solute atoms is given by:

$$\Delta G_f[X^q]=G_{tot}[X^q]-G_{tot}[bulk]-\sum_i n_i\mu_i^0\,, \tag{S.19}$$

where $G_{tot}[X^q]$ and $G_{tot}[bulk]$ are the Gibbs energies of defective and perfect structures, while n_i are are numbers of moles and μ are chemical potentials of the involved particles. The formation energy of a point defect can be used to determine its concentration at thermal equilibrium by computing its Boltzmann factor.

Mechanical properties

A large collection of data points taken within the harmonic region of bulk crystals were utilized in the optimization of our ReaxFF force field, in order to train it to describe the linear elastic stress-strain relationship in the Li–Si–C system that follow Hooke's law[17]

$$\sigma_i=\sum_{j=1}^{6} C_{ij}\varepsilon_j \text{ or } \varepsilon_i=\sum_{j=1}^{6} S_{ij}\sigma_j\,. \tag{S.20}$$

By considering the indices i and j as spatial directions {1,2,3,4,5,6}={xx,yy,zz,yz,xz,xy}, the stress elements σ_i and the strain elements ε_i and ε_j are linearly correlated according to the elastic constants C_{ij} or by the elements S_{ij} of the compliance tensor (*i.e.* inverse of the stiffness tensor C_{ij}) . In the elastic regime the strains are usually small enough so that the energy can be expanded harmonically as

$$E(\varepsilon_i,\varepsilon_j)=E(0,0)+\frac{1}{2}V_0\sum_{i=1}^{6}\sum_{j=1}^{6} C_{ij}\varepsilon_i\varepsilon_j\,, \tag{S.21}$$

with $E(0,0)$ and V_0 being the unstrained cell volume and energy, respectively. Le Page and Saxe[17,18] developed a least-square method for extracting elastic constants from formula (S.21). The averaging procedure developed by Voigt, Reuss and Hill[19,20] is used to determine both the bulk modulus:

$$B_H=\frac{B_V+B_R}{2}\text{, with } B_V=\frac{\overline{C_{11}}+2\overline{C_{12}}}{3} \text{ and } B_R=\frac{1}{3\overline{S_{11}}+6\overline{S_{12}}}\,, \tag{S.22}$$

as well as the shear modulus:

$$G_H=\frac{G_V+G_R}{2}\text{, with } G_V=\frac{\overline{C_{11}}-\overline{C_{12}}+3\overline{C_{44}}}{5} \text{ and } G_R=\frac{5}{4\overline{S_{11}}-4\overline{S_{12}}+3\overline{S_{44}}}\,. \tag{S.23}$$

The elastic constants $\overline{C_{ij}}$ for polycrystalline and cubically shaped systems are averaged from the corresponding C_{ij} elements. Further elastic properties can be derived from the bulk and shear moduli, such as Young's modulus

$$Y=\frac{9BG}{3B+G}, \tag{S.24}$$

the Poisson's ratio

$$\nu=\frac{3B-2G}{2(3B+G)}, \tag{S.25}$$

and Pugh's ratio[19,21]

$$k=\frac{B}{G}. \tag{S.26}$$

A material with a Pugh's ratio, k, below 1.75 is considered to be brittle. Above this threshold a material is considered to be ductile. Moreover, the Vickers hardness H_V is calculated using Tian's approach[22]:

$$H_V=0.92k^{1.137}G^{0.708}. \tag{S.27}$$

Thermal properties

The vibrational heat capacity due to phonons is:

$$C(T)=9NR\left(\frac{T}{\theta_D}\right)^3\int_{x=0}^{x=\theta_D/T}\frac{x^4e^x}{(e^x-1)^2}dx \tag{S.28}$$

and depends on the Debye temperature:

$$\theta_D=\frac{\hbar}{k_B}\left(\frac{6\pi^2N}{V_0}\right)^{1/3}v_m, \tag{S.29}$$

where v_m is the speed of sound depending on elastic constants[23]. N is the number of atoms in the cell whereas $\hbar$ and k_B are the reduced Planck and Boltzmann constants, respectively.

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